LLM Week 8 GA

1) Consider the following large language models:

1. BERT → E
2. GPT-1 → D
3. GPT-2 → D
4. T5 → ED
5. BART → ED
6. GPT-3 → D
7. GPT-4 → D
8. LLaMA → D
9. LLaMA-2 → D
10. Galactica → D

E → encoder
D → decoder
ED → enc – dec.

(With the above data, kindly answer the below Question No.1,2 and 3)

How many models from the list are encoder-only models?

Answer: ______

1

Yes, the answer is correct.
Score: 2

Accepted Answers:

(Type: Numeric) 1

2 points

2) How many models from the list are encoder-decoder models?

Answer: ______

2

Yes, the answer is correct.

3) How many models from the list are decoder-only models?

Answer: ______

7

Yes, the answer is correct.
Score: 2

Accepted Answers:

(Type: Numeric) 7

2 points

4) Choose all the aspects of the pre-training datasets that impact the model's performance ***2 points***

- [x] Quality
- [x] Size
- [x] Diversity
- [ ] None of these

$\left.\right\}$ 3 axes of dataset

5) Identify the correct scaling laws formula. The symbols have usual meaning ***2 points***

Answer: ______

- (x) $L(N,D) = \left[\left(\frac{N_c}{N}\right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D}\right]^{\alpha_D}$
- ( ) $L(N,D) = \left[\left(\frac{N_c}{N}\right)^{\frac{\alpha}{\alpha_N}} + \frac{D_c}{D}\right]^{\alpha_D}$
- ( ) $L(N,D) = \left[\left(\frac{N_c}{N}\right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D}\right]^{\alpha_N}$ α
- ( ) $L(N,D) = \left[\left(\frac{D_c}{D}\right)^{\frac{\alpha_N}{\alpha_D}} + \frac{N_c}{N}\right]^{\alpha_D}$ α
- ( ) None of these

$$\left(\left(\frac{N_c}{N}\right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D}\right)^{\alpha_D}$$

Yes, the answer is correct.
Score: 2

Accepted Answers:

$L(N,D) = \left[\left(\frac{N_c}{N}\right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D}\right]^{\alpha_D}$

6) Suppose you train a model with 1 billion parameters on a dataset with 1 billion tokens. What is the expected test loss by scaling law? Enter your answer correct upto 3 decimal places.

2.857

$$L = \left[\left(\frac{8.8 \times \cancel{10^{13}}\,10^4}{\cancel{1 \times 10^9}}\right)^{\frac{0.076}{0.095}} + \frac{5.4 \times \cancel{10^{13}}\,10^4}{\cancel{1 \times 10^9}}\right]^{0.095}$$

$$= \left[(88000)^{0.8} + 54000\right]^{0.095} = (63027)^{0.095} = 2.857$$

7) To get high quality content from raw data, what are the typical preprocessing steps involved? **2 points**

- [x] Removing duplicate content.
- [x] Removing toxic content.
- [x] Removing personally identifiable data.
- [x] Removing machine translated content.
- [x] Removing place holder text (e.g. " Lorem ipsum...")

Yes, the answer is correct.
Score: 2

Accepted Answers:

Removing duplicate content.
Removing toxic content.
Removing personally identifiable data.
Removing machine translated content.
Removing place holder text (e.g. " Lorem ipsum...")

8) Rajesh would like to reproduce the refined web pipeline and noted down the steps involved as following:
1. Line-wise correction.
2. Document-wise filtering.
3. Text extraction.
4. Language identification.
5. Repetition removal.
6. URL filtering.
7. Exact deduplication.
8. Fuzzy deduplication.

Enter the correct order of the steps in the pipeline.

63452187

Yes, the answer is correct.

9) Consider following datasets for pre-training an LLM:
1. BookCorpus 5 GB
2. Roots 5 TB
3. C4 → GB
4. WebText → less GB

Arrange them in decreasing order of their size in GB. Say, you com-pute the correct order to be "1234", then enter it without whitespaces or quotes.

2341

Yes, the answer is correct.
Score: 2

Accepted Answers:

(Type: Numeric) 2341

2 points

10) Select correct statements regarding pretraining dataset "Sangrah": **2 points**

- ( ) It is mono-lingual dataset. X multi
- (•) It has text from multiple Indian languages.
- ( ) It has no translated content from any other source. → it was with the help of Cy
- ( ) It has only English text. X